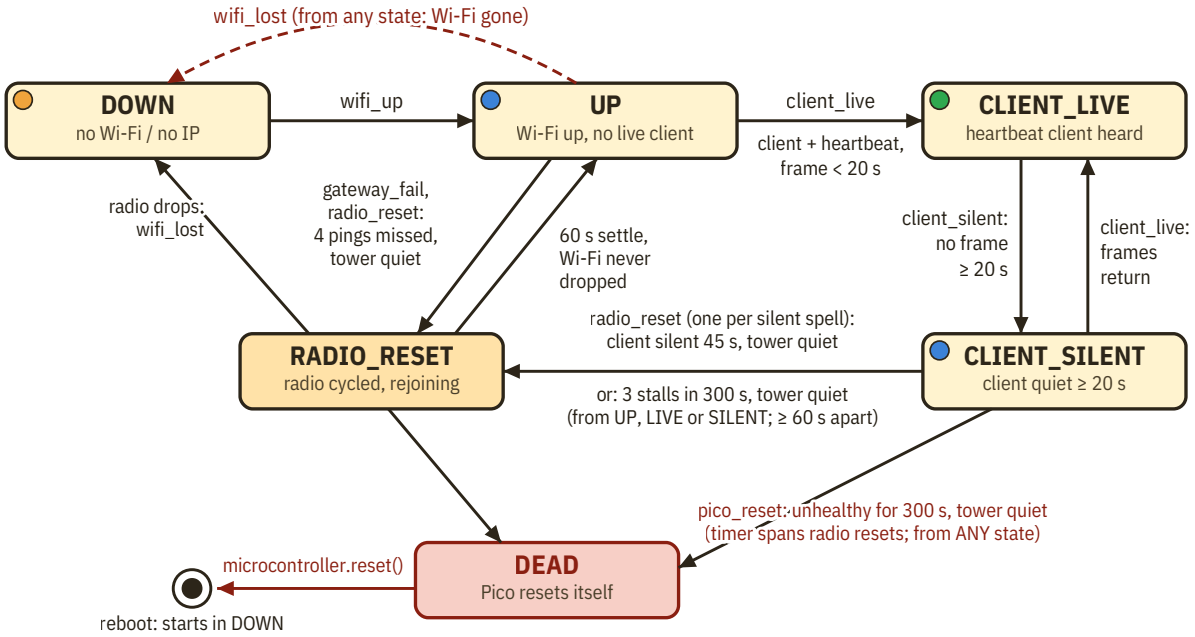
SANDSWING LINK STATE MODEL · as built

WI-FI AND CLIENT LINK (`LinkHealth.update()`, one pass every 2 s)



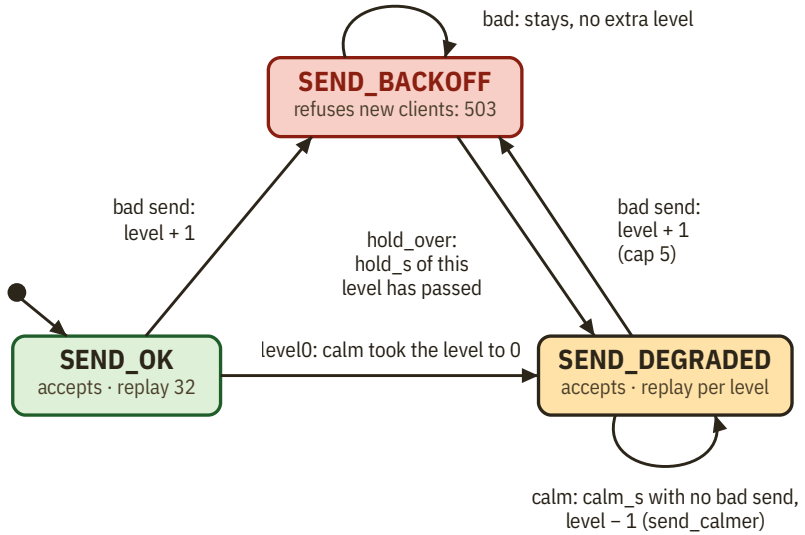
Dots = LED1 colour while in that state. LIVE and SILENT fall back to UP (no event) when clients go or no heartbeat has been heard this boot. "Tower quiet" = tower inactive and no stroke for 3 s. The **300 s** timer starts at the first bad sign (Wi-Fi down, gateway failing, client silent, 3 stalls) and runs across radio resets and rejoins; a working gateway ping, a live client, or 90 s of Wi-Fi up with no bad sign clears it. At most one Pico reset per spell. Every radio reset is logged as `link_dead`.

LED1 MEANINGS (Pico panel, `panel.LED_LINK`)

● **amber**	`no_wifi`: no Wi-Fi (clients dropped; rejoin every 30 s, 120 s while ringing)
● **blue**	`wifi_only`: Wi-Fi up, no live client
● **green**	`ws_client`: a heartbeat client heard within 20 s (before any client has sent a heartbeat this boot, any connected client counts)
● **red**	`error` colour exists in the table, but **nothing in the code sets it**: it is only the fallback for an unknown link value. (LED6 red is a different thing: never time-synced.)

Green is the Pico's own belief: a half-open link can still show green (known gap 1, page 2).

SEND PATH (`SendPath`, table `SEND_TRANSITIONS`)



level	hold	replay	frames/turn · gap	level falls after
0 (OK)	0	32	1 · 0.10 s	n/a
1	2 s	32	1 · 0.25 s	2 h without a bad send
2	4 s	16	1 · 0.50 s	2 h
3	8 s	8	1 · 0.50 s	1 h
4	16 s	4	1 · 1.00 s	30 min
5 (cap)	30 s	0	none	30 min

A **bad send** = failed, or slower than `SLOW_S`: replay 0.2, live 0.3, server 1.0, close 0.15, poll 1.5 s. Each bad send in OK or DEGRADED raises the level by one; bad sends during a hold add nothing. The calm timer counts from the latest of last bad send, level change and hold end, and is hours on purpose. A client connecting at level 3 or higher gets fewer than 32 old events (Sandbells shows a gap).

SANDSWING LINK STATE MODEL · as built

same sources and commits · page 2 of 2: the numbers, the log lines, the gaps, Sandbells, and what is only proposed

TIMING CONSTANTS (log from an incident log · src from firmware source · guess chosen, nothing measured)

PICO_HEARTBEAT_S	5 s	heartbeat both ways guess
LINK_CHECK_S	2 s	one supervision pass guess
CLIENT_SILENT_S	20 s	no frame: SILENT, client dropped guess (4 beats)
LINK_DEAD_S	45 s	silent spell to radio reset; was 120 s; a ~135 s degraded spell and a ~36 s heal were seen log
GATEWAY_CHECK_S / _FAIL_N	30 s / 4	ping only while quiet guess
STALL_S / _N / _WINDOW_S	5 s / 3 / 300 s	from a 12 min sick-radio incident with 10 s stalls log; the numbers guess
RADIO_RESET_SETTLE_S	60 s	reset heals in ~36 s log; the 60 guess
PICO_RESET_S	300 s	unhealthy spell to Pico reset guess
CALM_S	90 s	ends a spell guess
WIFI_REJOIN_S	30 / 120 s	quiet / ringing; a join blocks up to 10 s per network guess
POLL_BUDGET_S	1.0 s	one `/ws` poll guess
SEND_TIMEOUT_S	0.4 s	frame to one client; request / reply / frame bounds 0.3 s; close 0.1 s guess
hold, replay, calm	table	the ladder values are design choices guess
10 s inside one send()	200 × 50 ms	CircuitPython 10.0.3 lwIP retry; read from the source, **not measured on a Pico** src
PEER_SILENT_S	20 s	Sandbells ingest: no frame after a heartbeat, reconnect

LOG LINES TO GREP (`~/pico-serial.log`)

```
grep -E ' link (send_(backoff|degraded|calmer|ok)|client_(connected|dropped)|link_dead|wifi_(lost|back)|reset)|loop late|ws poll took' ~/pico-serial.log
```

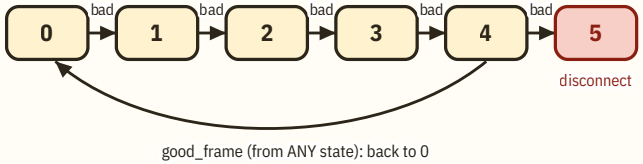
link send_backoff	bad send; refusing clients for `hold_s` (level, replay, why)
link send_degraded	hold over; gentler replay (level, refused)
link send_calmer	calm_s without a bad send; level − 1
link send_ok	back to level 0
link client_connected	client in (peer, replay size)
link client_dropped	client out (reason, hb, up_s)
link link_dead	a **radio reset is being requested** (silence, 4 gateway misses or 3 stalls), then the radio is cycled
link wifi_lost	Wi-Fi gone (count)
link wifi_back	rejoined (ip, down_s, rejoins)
link reset	300 s spell: the Pico resets itself (DEAD)

A healthy Pico prints no `send_*` lines and no `ws poll took`.

KNOWN GAPS

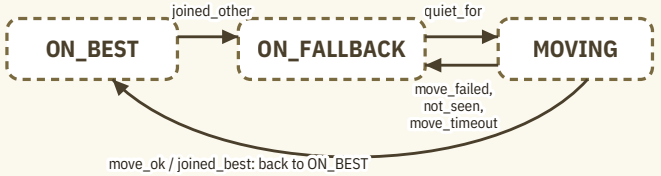
1. **Wedge seen 2026-10-28 18:27-18:37:** Pico LED green, loop alive, unreachable ("No route to host") for ~9 min; the 300 s reset did not recover it. Cause **unconfirmed**: the serial log was off (Pico on a power bank).
2. CircuitPython 10.0.3 `send()` can block ~10 s inside lwIP; Python deadlines are only checked after it returns.
3. The Pico cannot read its rssi (`ap_info` fails), so no signal-based decisions.
4. The gateway ping is sampled only while the tower is quiet.

SANDBELLS INGEST: BAD FRAMES (`framestate.py`, branch `ingest-bad-frame`)



State = bad frames in a row; events `good_frame` / `bad_frame`. `bad_frame`: count, skip, one WARNING per 10 s; the **5th in a row** adds *disconnect* (`BAD_FRAME_LIMIT` 5, a first guess to be replaced from the journal). `good_frame` (any frame that decodes, even a heartbeat Sandbells ignores) resets; a new connection starts at 0.

PROPOSED, NOT DEPLOYED NETPREF: PREFER THE FIRST WI-FI NETWORK



Separate commit, not in the model on page 1. Move off a fallback network when quiet (tower inactive, no stroke for 60 s, joined for 60 s); after a failure try again after 5, 10, 20, then 30 min. Log lines `net_prefer`, `net_moved`, `net_stay`. Every number is a first guess; the hotspot `Sandbells` is never moved from or to.

PROPOSED, NOT BUILT NEW EVENT: SILENCE OF EVERY KIND



Event: **no inbound traffic of any kind for N s while a client is believed live** (N not chosen). It would feed the **same reset path** as the other unhealthy states: bad sign → radio reset → 300 s unhealthy spell → Pico reset. Motivation: known gap 1 (green but unreachable). Which inbound signals count is still to be decided.

WHERE THE DOCS AND THE CODE DIFFER

- Only `SendPath` is a transition table; the Wi-Fi/client machine is if-chains in `update()`.
- `docs/link-health.md` draws DEAD from DOWN and SILENT only; the code reaches it from **any** state.
- With a client still connected, the 45 s counts from entering SILENT (20 s after its last frame), so about 65 s after it.
- `link_dead` is logged for **every** radio reset, not only silence.